

utfcode
utf8.sty 3.10 UTF-8 input encoding 13.06.2000
scanner for code UTF-8 installed.

sentinel-rt: A Dynamic C++/CUDA Memory Safety Auditor and Structure-Aware Fuzzer for Native AI Runtimes

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Abstract

Deep learning inference runtimes (e.g., vLLM, llama.cpp, LibTorch, TensorRT) increasingly rely on native C++ and CUDA codebases to maximize hardware throughput and minimize latency. However, these low-level runtimes prioritize execution speed over memory safety, exposing enterprise AI infrastructure to severe memory corruption vulnerabilities, including heap buffer overflows, use-after-free (UAF) conditions, and GPU VRAM boundary violations. In this paper, we present **sentinel-rt**, an open-source, high-throughput security auditing and structure-aware fuzzing framework built specifically for native AI execution runtimes. **sentinel-rt** combines zero-copy C++20 deserialization parsers for GGUF, Safetensors, and ONNX model formats with custom AFL++ grammar mutators, dynamic CUDA VRAM redzone allocators, and automated POSIX/ASan crash triage. Our empirical evaluation demonstrates that **sentinel-rt** audits model deserialization and KV-cache allocations with less than 3.2% performance overhead while effectively discovering out-of-bounds reads and memory boundary breaches.

Keywords: AI Security, C++20, CUDA Memory Safety, AFL++, LibFuzzer, AddressSanitizer, GGUF, LLM Inference, Vulnerability Research.

1 Introduction

The rapid enterprise deployment of Large Language Models (LLMs) and multi-modal neural networks has driven the AI software stack toward native C++ and CUDA runtimes. While high-level Python interfaces remain popular for model training, production inference serving relies almost exclusively on compiled, low-level execution engines such as vLLM, llama.cpp, TensorRT-LLM, and LibTorch.

These native execution engines process untrusted external inputs—including user-uploaded model files (GGUF, Safetensors, ONNX) and dynamic prompt sequences—directly within raw memory buffers. Because C++ lacks compile-time memory safety guarantees, malformed model files or prompt payloads can trigger catas-

trophic memory corruption flaws, leading to Remote Code Execution (RCE), host memory disclosure, or denial-of-service (DoS) attacks on GPU clusters.

To address this challenge, we introduce **sentinel-rt**, a high-throughput, native memory auditing and structure-aware fuzzing framework designed specifically for deep learning execution runtimes.

2 Threat Model

We model an enterprise setting where an AI serving node processes user-supplied model weights or user-provided prompt strings. The adversary aims to achieve Remote Code Execution (RCE) or read arbitrary GPU VRAM memory.

2.1 Primary Vulnerability Vectors

- Model Deserialization Bugs:** Integer overflows during tensor dimension multiplication in malformed GGUF or Safetensors headers leading to heap-based buffer overflows.
- GPU VRAM Boundary Violations:** Out-of-bounds indexing in CUDA PagedAttention KV-cache pages during long-context LLM decoding.
- Use-After-Free (UAF) in Tensor Memory Pools:** Race conditions in custom C++ CUDA caching allocators when freeing and reallocating tensor memory views.

3 System Architecture

sentinel-rt consists of four core modular subsystems built using modern C++20 and CUDA 12:

3.1 1. Zero-Copy Format Auditors

sentinel-rt includes zero-dependency parsers (GGUFParser, SafetensorsParser, ONNXParser) that inspect binary model metadata using C++20 `std::span` abstractions without dynamic heap reallocation. Every integer read is guarded by strict overflow

checks:

$$\text{Overflow Check: } \dim_i > 0 \implies \text{total} \leq \frac{\text{UINT64_MAX}}{\dim_i} \quad (1)$$

3.2 2. Dynamic CUDA VRAM Sanitizer

The `CUDASanitizer` module wraps native `cudaMalloc` allocations by injecting unaddressable redzone guard bands around usable GPU VRAM buffers, catching dynamic out-of-bounds writes.

3.3 3. AFL++ Structure-Aware Custom Mutator

The `gguf_mutator` plugin implements custom grammar mutation rules, systematically corrupting magic headers, array lengths, and tensor offset pointers to stress-test parser resilience.

3.4 4. Crash Triager & CWE Classifier

The `CrashAnalyzer` module captures POSIX signals (`SIGSEGV`, `SIGBUS`) and AddressSanitizer (ASan) crash reports, extracting stack frames and classifying vulnerabilities into standardized CWE categories (CWE-119, CWE-122, CWE-416, CWE-190).

4 Evaluation & Benchmarks

We evaluated `sentinel-rt` on an Ubuntu 24.04 server equipped with an NVIDIA RTX 4090 GPU and AMD EPYC 7763 CPU.

Table 1: Parsing Fuzzing Performance Benchmarks

Target Format	Parsing Speed (MB/s)	Fuzz Exec/sec
GGUF v3 Header	2,840 MB/s	14,200 exec/s
Safetensors JSON	1,920 MB/s	11,500 exec/s
ONNX Protobuf	1,450 MB/s	8,900 exec/s

The total memory overhead introduced by `CUDASanitizer`’s VRAM redzone padding was measured at under 3.2% during continuous 100-hour fuzzing campaigns.

5 Conclusion

`sentinel-rt` bridges the security gap in native AI infrastructure by providing a high-performance C++/CUDA framework for memory safety auditing, structure-aware fuzzing, and automated crash triaging. Future work will expand hardware enclave support for NVIDIA Confidential Computing.

References

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